Chapter 16
Materials for optical devices





Instruments using the ability of materials to reflect and refract. (Image of microscope courtesy of A–Z Microscope Corporation, California USA.)

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16.1 Introduction and synopsis

It was at one time thought that the fact that light could travel through space—from the sun to earth, for instance—must mean that space was not really empty but filled with 'luminiferous ether'. It was not until the experiments of Michelson[1] and Morley in 1881 that it was realized that light did not need a 'material' for its propagation but could propagate through totally empty space at what is now seen as the ultimate velocity: 3×10^8m/s.

When radiation strikes materials, things can happen. Materials interact with radiation by reflecting it, absorbing it, transmitting it and refracting it. This chapter is about these interactions, the materials that do them best and the ways we use them. The chapter opening page shows two: a reflecting telescope and a refracting microscope, each of which depend on the optical properties of materials.

16.2 The interaction of materials and radiation

Electromagnetic (e-m) radiation permeates the entire universe. Observe the sky with your eye and you see the visible spectrum, the range of wavelengths we call 'light' (0.40–0.77 μm). Observe it with a detector of X-rays or γ-rays and you see radiation with far shorter wavelengths (as short as 10^{-4}nm, one-thousandth the size of an atom). Observe it instead with a radio-telescope and you pick up radiation with wavelengths measured in millimeters, meters or even kilometers, known as radio and microwaves. The range of wavelengths of radiation is vast, spanning 18 orders of magnitude (Figure 16.1). The visible part of this spectrum is only a tiny part of it—but even that has entrancing variety, giving us colors ranging from deep purple through blue, green and yellow to deep red.

The intensity I of an e-m wave, proportional to the square of its amplitude, is a measure of the energy it carries. When radiation with intensity I_o strikes a material, a part I_R of it is reflected, a part I_A absorbed and a part I_T may be transmitted. Conservation of energy requires that

$$\frac{I_R}{I_o} + \frac{I_A}{I_o} + \frac{I_T}{I_o} = 1 \tag{16.1}$$

The first term is called the *reflectivity* of the material, the second the *absorptivity* and the last the *transmittability* (all dimensionless). Each depends on the wavelength of the radiation, on the nature of the material and on the state of

[1] Albert A. Michelson (1852–1931), Prussian-American experimental physicist, who, with E.W. Morley, first demonstrated that the speed of light is independent of the earth's motion, a finding central to the establishment of the theory of relativity.

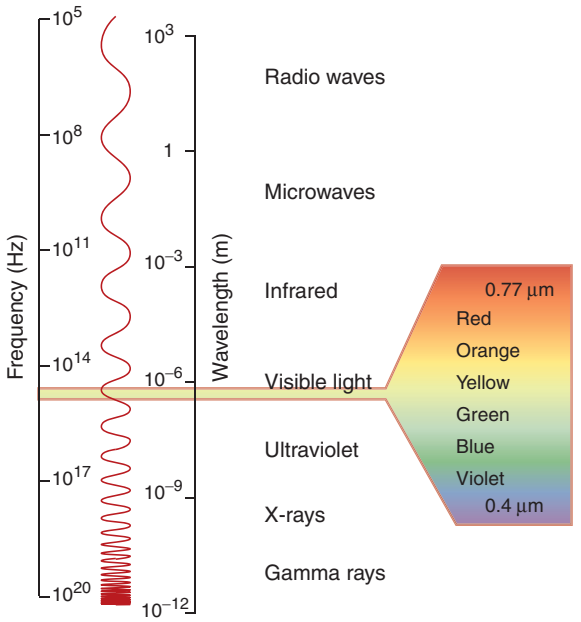


Figure 16.1 The spectrum of electromagnetic (e-m) waves. The visible spectrum lies between the wavelengths 0.4 and 0.77 μm.

its surfaces. They can be thought of as properties of the material in a given state of surface polish, smoothness or roughness.

In optics we are concerned with wavelengths in the visible spectrum. Materials that reflect or absorb all visible light, transmitting none, are called *opaque*, even though they may transmit in the near visible (infrared or ultraviolet). Those that transmit a little diffuse light are called *translucent*. Those that transmit light sufficiently well that you can see through them are called *transparent*; a subset of these that transmit almost perfectly, making them suitable for lenses, light-guides and optical fibers, are given the additional title of *optical quality*. Metals are opaque. To be transparent a material must be a dielectric.

Specular and diffuse reflection

Metals reflect almost all the light that strikes them; none is transmitted and little is absorbed. When light strikes a reflecting surface at an incident angle θ_1, part of it is reflected, leaving the surface with an angle of reflection θ_2 such that

$$\theta_1 = \theta_2 \tag{16.2}$$

Specular surfaces are microscopically smooth and flat. A beam striking such a surface suffers specular reflection, meaning that it is reflected as a beam, as on

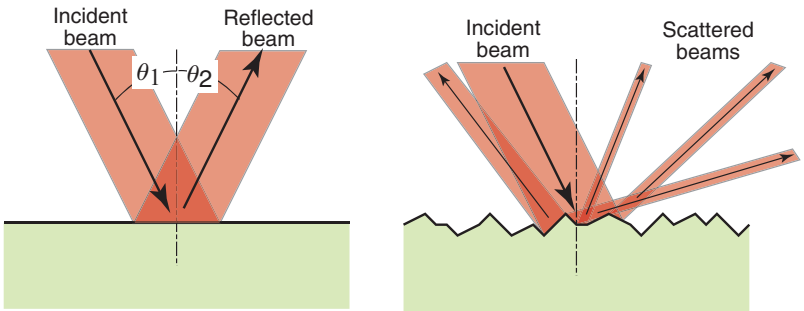


Figure 16.2 Optically flat, reflective surfaces give specular reflection, such that $\theta_1 = \theta_2$. The angles of incidence and reflection are always equal, but the rough surface gives diffuse reflection even though the angles of incidence and reflection are still, locally, equal.

the left of Figure 16.2. *Diffuse* surfaces are irregular; the law of reflection (equation (16.2)) still holds locally but the incident beam is reflected in many different directions because of the irregularities, as on the right of the figure.

Absorption

If radiation can penetrate a material, some is absorbed. The greater the thickness x through which the radiation passes, the greater the absorption. The intensity I, starting with the initial value I_0, decreases such that

$$I = I_o \exp - \beta x \tag{16.3}$$

where β is the absorption coefficient, with dimensions of m^{-1} (or, more conveniently, mm^{-1}). The absorption coefficient depends on wavelength, with the result that white light passing through a material may emerge with a color corresponding to the wavelength that is least absorbed—that is why a thick slab of ice looks blue.

Transmission

By the time a beam of light has passed completely through a slab of material it has lost some intensity through reflection at the surface at which it entered, some in reflection at the surface at which it leaves, and some by absorption in between. Its intensity is

$$I = I_o \left(1 - \frac{I_R}{I_o}\right)^2 \exp - \beta x \tag{16.4}$$

The term $(1 - I_R/I_o)$ occurs to the second power because intensity is lost through reflection at both surfaces.

Refraction

The velocity of light in a vacuum, $c_o = 3 \times 10^8$m/s, is as fast as it ever goes. When it (or any other electromagnetic radiation) enters a material, it slows down. The *index of refraction*, n, is the ratio of its velocity in a vacuum, c_o, to that in the material, c:

$$n = \frac{c_o}{c} \tag{16.5}$$

This retardation makes a beam of light bend or *refract* when it enters a material of different refractive index. When a beam passes from a material 1 of refractive index n_1 into a material 2 of index n_2 with an angle of incidence θ_1, it deflects to an angle θ_2, such that

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{n_2}{n_1} \tag{16.6}$$

as in Figure 16.3(a); the equation is known as Snell's law[2]. The refractive index depends on wavelength, so each of the colors that make up white light is diffracted through a slightly different angle, producing a spectrum when light passes through a prism. When material 1 is vacuum or air, for which $n_1 = 1$, the equation reduces to

$$\frac{\sin \theta_1}{\sin \theta_2} = n_2$$

Equation (16.6) says that light passes from a material with index $n_1 = n$ into air with $n_2 = 1$, it is bent away from the normal to the surface, like that in Figure 16.3(b). If the incident angle, here θ_1, is slowly increased, the emerging beam tips down until it becomes parallel with the surface when $\theta_2 = 90°$ and $\sin \theta_2 = 1$. This occurs at an incident angle, from equation (16.6), of

$$\sin \theta_1 = \frac{1}{n} \tag{16.7}$$

For values of θ_1 greater than this, the equation predicts values of $\sin \theta_2$ that are greater than 1, and that can't be. Something strange has to happen, and it does: the ray is totally reflected back into the material, as in Figure 16.3(c). This *total*

[2]Willebrord Snell (1591–1626), Dutch astronomer, also known as Snell van Royen, or Snellius, who first derived the relationship between the different angles of light as it passes from one transparent medium to another. The lunar crater Snellius is named after him.

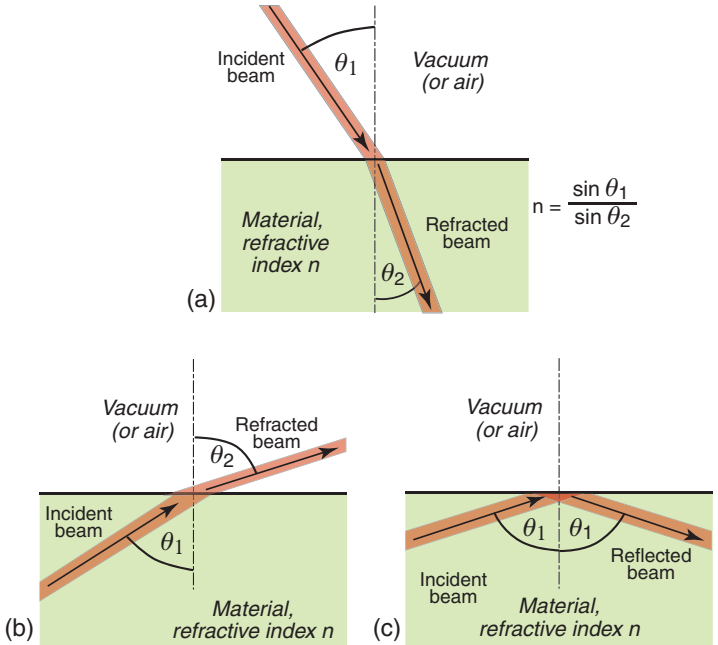


Figure 16.3 Refraction and total internal reflection.

internal reflection has many uses, one being to bend the path of light in prismatic binoculars and reflex cameras. Another is to trap light within an optical fiber, an application that has changed the way we communicate.

Reflection is related to refraction. When light traveling in a material of refractive index n_1 is incident normal to the surface of a second material with a refractive index n_2, the reflectivity is

$$R = \frac{I_R}{I_o} = \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 \tag{16.8a}$$

If the incident beam is in air, with refractive index 1, this becomes

$$R = \frac{I_R}{I_o} = \left(\frac{n - 1}{n + 1}\right)^2 \tag{16.8b}$$

Thus, materials with high refractive index have high reflectivity.

Before leaving Snell and his law, ponder for a moment on the odd fact that the beam is bent at all. Light entering a dielectric, as we have said, slows down.

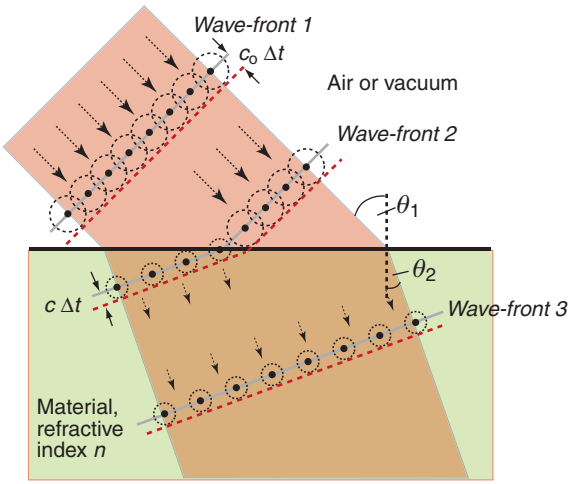


Figure 16.4 The Fresnel construction, explaining why a light beam is bent on entering a material of higher refractive index.

If that is so why does it not continue in a straight line, just more slowly? Why should it bend? To understand this we need Fresnel's[3] construction. We think of light as advancing via a series of wave-fronts. Every point on a wave-front acts as a source, so that, in a time Δt the front advances by $c\Delta t$, where c is the velocity of light in the medium through which it is passing ($c = c_o$ in vacuum or air)—as shown in Figure 16.4 with *wave-front 1* advancing $c_o\Delta t$. When the wave enters a medium of higher refractive index it slows down so that the advance of the wave-front within the medium is less than that outside, as shown with *wave-front* 2 advancing $c\Delta t$ in the figure. If the angle of incidence θ_1 is not zero, the wave-front enters the second medium progressively, causing it to bend, so that when it is fully in the material it is traveling in a new direction, characterized by the angle of refraction, θ_2. Simple geometry then gives equation (16.6).

16.3 Charts for optical properties

Refractive index and dielectric constant

Figure 16.5 shows the refractive index, n, of dielectrics, plotted against the dielectric constant ε_R (defined in Chapter 14). Those shown as white bubbles are transparent to visible light; those with red bubbles are not, but transmit

[3] Augustin Jean Fresnel (1788–1827), French physicist and engineer, known for his research on the wave theory of light and of diffraction and polarization.

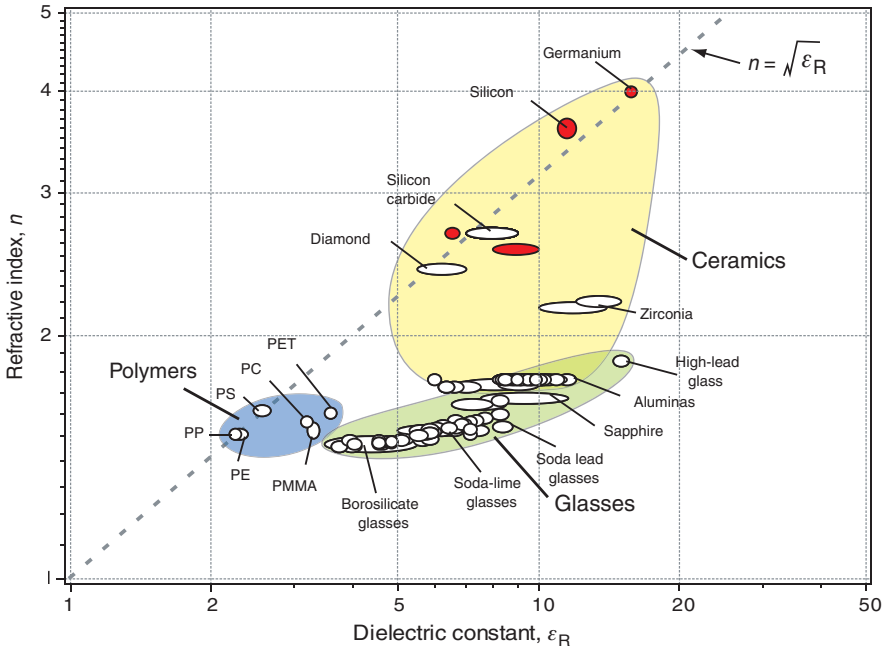


Figure 16.5 A chart of refractive index and dielectric constant, showing the approximate relationship $n = \sqrt{\varepsilon_R}$. (This chart and the next were made using Level 3 of the CES database.)

radiation of longer wavelengths (germanium, for instance, is used for lenses for infrared imaging). Note the wide and continuous range of refractive index of glasses, determined by their chemistry. This ability to control n by manipulating composition is central to the selection of materials for lenses and optical fibers. The relationship between n and ε_R is explored in Section 16.4.

Reflectivity and refractive index

The spectacular refraction and sparkle of diamond comes from its high refractive index, giving reflectance and extensive total internal reflection. Figure 16.6 plots these two properties. As in the previous chart, transparent materials are shown as white circles, those that are opaque as red. As the reflectance depends on n, the materials fall on a characteristic curve. High refractive index gives high reflectivity, and it is here that diamond ($n = 2.42$) excels; this, its unsurpassed hardness and durability ('diamonds are forever') and its scarcity create its unique desirability as jewelry. Costume jewelry uses cheap alternatives that mimic diamond in having high refractive index: zircon, $ZrSiO_4$ ($n = 1.98$) and cubic zirconia, ZrO_2 ($n = 2.17$). Plastic jewelry can't come close; as the chart shows, polystyrene (PS), with $n = 1.58$, has the highest refractive index among

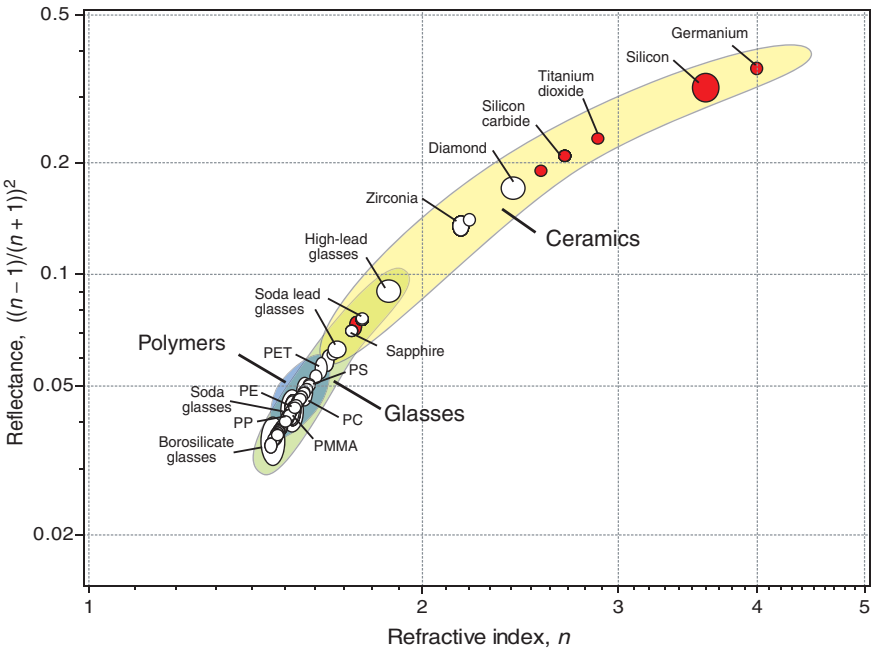


Figure 16.6 A chart of refractive index and reflectance. The materials with open circles are transparent to visible light; those with red circles are transparent to infrared and below, but not to visible radiation. Diamond is special because of its high refractive index, reflectance and transparency in the visible spectrum.

polymers (one reason it is used for the so-called 'jewel' cases in which CDs are packaged) but it is much lower than those of real jewels.

16.4 Drilling down: the physics and manipulation of optical properties

Light, like all radiation, is an electromagnetic (e-m) wave. The coupled fields are sketched in Figure 16.7. The electric part fluctuates with a frequency ν that determines where it lies in the spectrum of Figure 16.1. A fluctuating electric field induces a fluctuating magnetic field that is exactly $\pi/2$ out of phase with the electric one, because the induction is at its maximum when the electric field is changing most rapidly. A plane-polarized beam looks like this one: the electric and magnetic fields lie in fixed planes. Natural light is not polarized; then the wave also rotates so that the plane containing each wave continuously changes.

Many aspects of radiation are most easily understood by thinking of it as a wave. Others need a different picture—that of radiation as discrete packets of

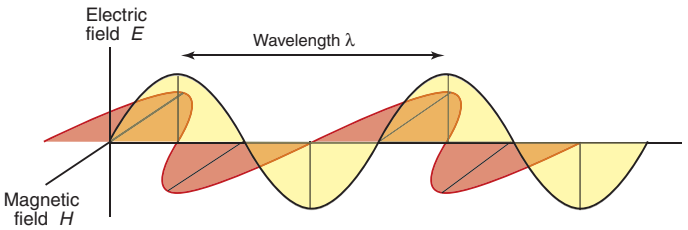


Figure 16.7 An electromagnetic wave. The electric component (gray) is $\pi/2$ out of phase with the magnetic component (red).

energy, *photons*. The idea of a wave that is also discrete energy units is not intuitive—it is another of the results of quantum theory that do not correspond to ordinary experience. The energy E_{ph} of a photon of radiation of frequency ν or wavelength λ is

$$E_{ph} = h\nu = \frac{hc}{\lambda} \tag{16.9}$$

where h is Planck's[4] constant (6.626×10^{-34} J.s) and c is the speed of the radiation. Thus, radiation of a given frequency has photons of fixed energy, regardless of its intensity—an intense beam simply has more of them. This is the key to understanding reflection, absorption and transmission.

Why aren't metals transparent?

Recall from Chapter 14 that the electrons in materials circle their parent atom in orbits with discrete energy levels, and only two can occupy the same level. Metals have an enormous number of very closely spaced levels in their conduction band; the electrons in the metal only fill part of this number. Filling the levels in a metal is like pouring water into a container until it is part full—its surface is the Fermi[5] level; levels above it are empty. If you 'excite' the water—say, by shaking the container—some of it can slosh to a higher level. If you stop sloshing, it will return to its Fermi level.

Radiation excites electrons; and in metals there are plenty of empty levels in the conduction band into which they can be excited. But here quantum effects cut in. A photon with energy $h\nu$ can excite an electron only if there is an energy level that is exactly $h\nu$ above the Fermi level—and in metals there is. So all the photons of a light beam are captured by electrons of a metal, regardless of their wavelength. Figure 16.8 shows, on the left, what happens to just one.

[4] Max Karl Ernst Ludwig Planck (1858–1947), a central figure in the development of quantum theory; it was he who formulated equation (16.9).

[5] Enrico Fermi (1901–1954), devisor of the statistical laws known as Fermi statistics governing the behavior of electrons in solids. He was one of the leaders of the team of physicists on the Manhattan Project for the development of the atomic bomb.

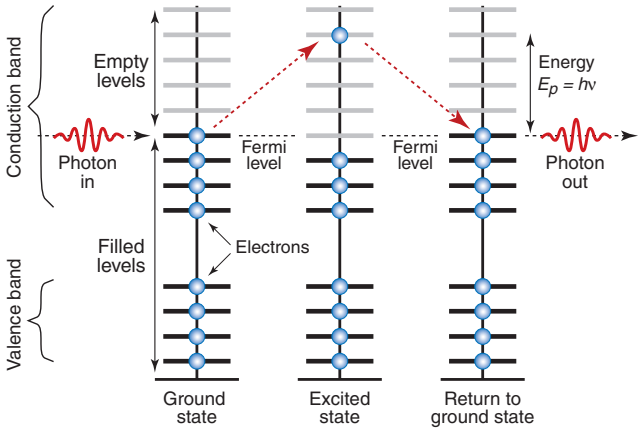


Figure 16.8 Metals absorb photons, capturing their energy by promoting an electron from the filled part of the conduction band into a higher, empty level. When the electron falls back, a photon is re-emitted.

What next? Shaken water settles back, and electrons do the same. In doing so it releases a photon with exactly the same energy that excited it in the first place, but in a random direction. Any photons moving into the material are immediately recaptured, so none make it more than about 0.01 μm (about 30 atom diameters) below the surface. All, ultimately, re-emerge from the metal surface—that is, they are reflected. Many metals—silver, aluminum and stainless steel are examples—reflect all wavelengths almost equally well, so if exposed to white light they appear silver. Others—copper, brass, bronze—reflect some colors better than others and appear colored because, in penetrating this tiny distance into the surface, some wavelengths are slightly absorbed.

Reflection by metals, then, has to do with electrons at the top of the conduction band. These same electrons provide electrical conduction. For this reason, the best metallic reflectors are the metals with the highest electrical conductivities—the reason that high-quality mirrors use silver and cheaper ones use aluminum.

How does light get through dielectrics?

If electrons snatch up photons, how is that some materials are transparent—light goes straight through them? Non-metals interact with radiation in a different way. Part may be reflected but much enters the material, inducing both dielectric and magnetic responses. Not surprisingly, then, the velocity of an e-m wave depends on the dielectric and magnetic properties of the material through which it travels. In a vacuum the velocity is

$$c_o = \frac{1}{\sqrt{\varepsilon_o \mu_o}} \tag{16.10}$$

where ε_o is the electric permittivity of a vacuum and μ_o its magnetic permeability, defined in Chapters 14 and 15. Within a material its velocity is

$$c = \frac{1}{\sqrt{\varepsilon\,\mu}} \tag{16.11}$$

where $\varepsilon = \varepsilon_R\ \varepsilon_o$ and $\mu = \mu_R\ \mu_o$ are the permittivity and permeability of the material. The refractive index, therefore, is

$$n = \frac{c_o}{c} = \sqrt{\varepsilon_R\ \mu_R} \tag{16.12}$$

The relative permeability μ_R of most dielectrics is very close to unity—only the magnetic materials of Chapter 15 have larger values. Thus,

$$n \approx \sqrt{\varepsilon_R} \tag{16.13}$$

The chart of Figure 16.5 has n as one axis and ε_R as the other. The diagonal line is a plot of this equation (the log scales make it a straight line). Polymers and very pure materials like diamond, silicon and germanium lie close to the line but the agreement with the rest is not so good. This is because refractive index and dielectric constant depend on frequency. Refractive index is usually measured optically, and that means optical frequencies, around 10^{15} Hz. Dielectric constants are more usually measured at radio frequencies or below—10^6 Hz or less. If the two are measured at the same frequency, equation (16.13) holds.

The reason that radiation of certain wavelengths can enter a dielectric is that its Fermi level lies at the top of the valence band, just below a band gap (Chapter 14 and Figure 16.9). The conduction band, with its vast number of empty levels, lies above it. To excite an electron across the gap requires a photon with an energy at least as great as the width of the gap, ΔE_{gap}. Thus, radiation with photon energy less than ΔE_{gap} cannot excite electrons—there are no energy states within the gap for the electron to be excited into. The radiation sees the material as transparent, offering no interaction of any sort, so it goes straight through.

Electrons are, however, excited by radiation with photons energies greater than ΔE_{gap} (i.e. higher frequency, shorter wavelength). These have enough energy to pop electrons into the conduction band, leaving a 'hole' in the valence band from which they came. When they jump back, filling the hole, they emit radiation of the same wavelength that first excited them, and for these wavelengths the material is not transparent (Figure 16.9, right-hand side). The critical frequency ν_{crit}, above which interaction starts, is given by

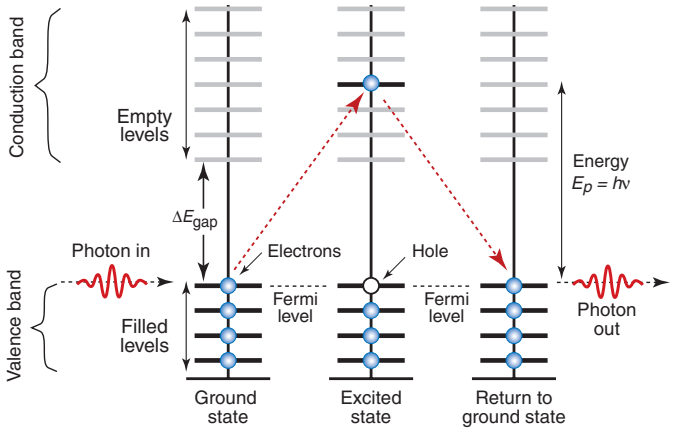


Figure 16.9 Dielectrics have a full band, separated from the empty conduction band by an energy gap. The material cannot capture photons with energy less than ΔE_{gap}, meaning that, for those frequencies, the material is transparent. Photons with energy greater than ΔE_{gap} are absorbed, as illustrated here.

$$h\nu_{crit} = \Delta E_{gap} \tag{16.14}$$

The material is opaque to frequencies higher than this. Thus, bakelite is transparent to infrared light because its frequency is too low and its photons too feeble to kick electrons across the band gap, but the visible spectrum has higher frequencies with more energetic photons, exceeding the band gap energy; they are captured and reflected.

Although dielectrics can't absorb radiation with photons of energy less than that of the band gap, they are not all transparent. Most are polycrystalline and have a refractive index that depends on direction; then light is *scattered* as it passes from one crystal to another. Imperfections, particularly porosity, do the same. Scattering, sketched in Figure 16.10, makes the material appear translucent or even opaque, even though, when made as a perfect single crystal, it is completely transparent. Thus, sapphire (alumina, Al_2O_3), used for watch crystals and cockpit windows of aircraft, is transparent, but the polycrystalline, slightly porous form of the same material that is used for electronic substrates is translucent or opaque. Scattering explains why some polymers are translucent or opaque: their microstructure is a mix of crystalline and amorphous regions with different refractive indices. It explains, too, why some go white when you bend them: it is because light is scattered from internal microcracks—the crazes described in Chapter 6.

Color

If a material has a band gap with an energy ΔE_{gap} that lies within the visible spectrum, the wavelengths with energy greater than this are absorbed and those